Informativity Expectations Affect the Evaluation of Reasons

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Abstract

This dissertation pursues a pragmatic analysis of the "over-explanation" phenomenon. On standard probabilistic accounts of argument strength (e.g. Oaksford and Hahn, 2004), the posterior probability of a given conclusion monotonically increases when more evidence (in the form of *reasons*) is provided. This paper develops an alternative analysis based on information-theoretic models of discourse interpretation (e.g. Kravtchenko and Demberg, 2022a). If listeners expect rational speakers to produce efficient utterances, and listeners modify their background beliefs to rationalize perceived redundancies, then providing too much explanation can negatively influence the listener's posterior belief in a proposition, if he infers that the speaker expected him to disagree with her. The model is developed using the Rational Speech Act (RSA; Frank and Goodman, 2012) framework, and then tested via a behavioral experiment. The results of the experiment offer evidence in favor of the pragmatic analysis, demonstrating that pragmatic reasoning has a role to play in the truth-evaluation of semi-cooperative utterances.

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Chapter 1

Introduction

Language is built upon a foundation of trust, and there is a case to be made that this trust is rational. Our modern lives depend almost entirely on knowledge we acquire from others (Levy, 2021; Mercier and Sperber, 2017), and language use is evolutionarily stable only because cooperativity is generally advantageous (McCready, 2015). However, listeners obviously do not always accept assertions automatically on the basis of conversational norms (Farkas and Bruce, 2010; Sperber et al, 2010), and speakers do not expect their addressees to take implausible testimony for granted (Oey et al, 2023). Even when Grice's (1975) Quality maxim is presumably in effect, interlocutors might have access to different bodies of evidence against which they evaluate facts, or else they might differ subjectively on matters of taste or opinion. This study concerns the use of *reasons* to negotiate the acceptance of controversial beliefs.

In argumentative scenarios, the function of a reason is to reduce or eliminate the need for trust among interlocutors by instead making the target content more easily inferrable from the listener's accepted beliefs (Mercier, 2020). Beliefs justified by many coherent reasons, then, should be more credible– or at least, no less credible– than beliefs justified by few or none. This analysis is intuitive, and it is predicted by a number of formal and probabilistic models of argument strength (e.g. Oaksford and Hahn, 2004; Godden and Zenker, 2018). The present study, however, will investigate the competing intuition that arguments are actually weakened when they are justified redundantly, that is, when (far) more explanation is provided than what would be expected or required to raise an interlocutor's credence to the threshold for acceptance. Put famously by Queen Gertrude in Act III, Scene II of Hamlet: "The lady doth protest too much, methinks."

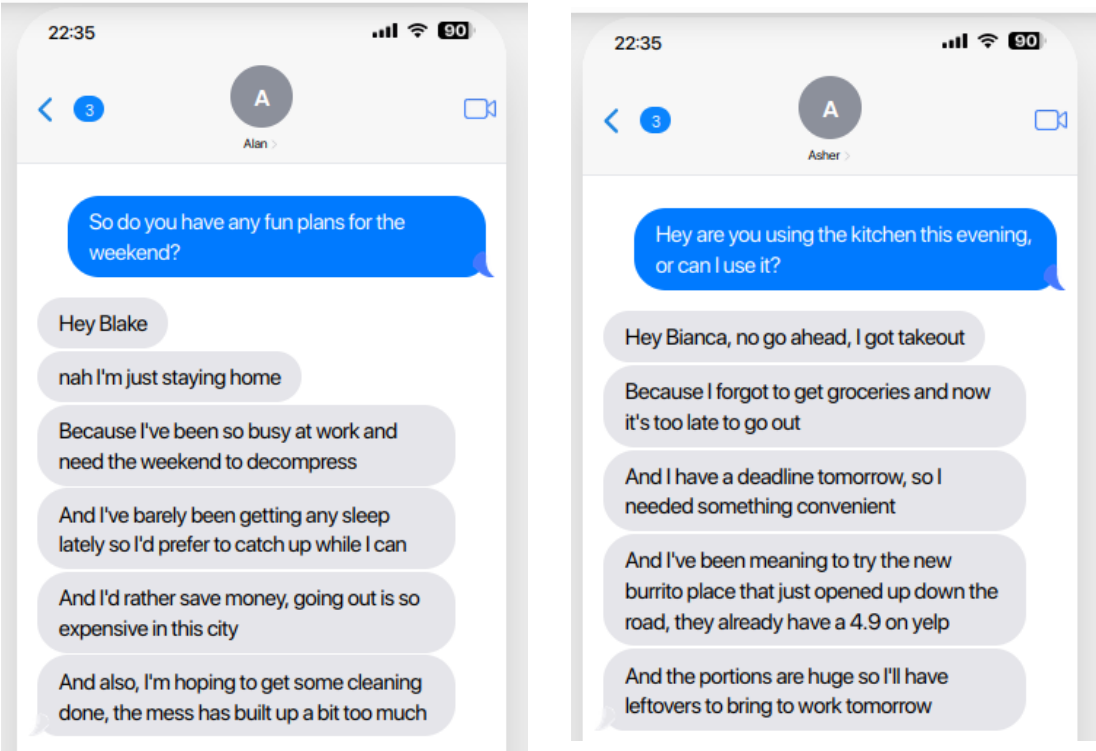


Figure 1.1: Examples of over-explanation.

This paper proposes a pragmatic account of this effect, based on computational and psycholinguistic theories of informativity. Briefly, if speakers are to make *efficient* usage of their cognitive resources to maximize information exchange while avoiding misunderstandings, then longer signals should be reserved for meanings that are more surprising, or less predictable (Levy and Jaeger, 2007 a.m.o). Comprehenders correspondingly interpret utterances with an expectation for informativity (Rohde and Rubio-Fernandez, 2022; Sedivy, 2003). They assume that rational speakers do not mention predictable content, and they derive pragmatic inferences to rationalize apparent inefficiencies. For example, assertion of trivial script continuations can induce atypicality inferences, e.g. "She went to the grocery store, and **she paid the cashier!**" generates an inference that the subject might not normally pay for her groceries (Kravtchenko and Demberg, 2022a; 2022b).

Most prior work on informativity has been concerned with pragmatic *interpretation*. When the Quality maxim can be taken for granted, the listener is expected to recover the speaker's intended meaning and update his beliefs by default. This work extends the informativity literature to the *evaluation* of arguments, where listeners choose to update their beliefs based on the soundness of the evidence on offer (Grice, 1969, Sperber et al,

2010). Pragmatic reasoning can account for the "over-explanation" effect if listeners evaluate evidence in light of an efficient speaker's choice to provide it.

More specifically, since assertions are typically accepted on the basis of conversational norms, a rational speaker would only expend effort to justify a belief if she expected the listener might challenge it otherwise. Further, she would provide sufficient evidence to ensure admission, with no superfluous reasons. So in the case of over-explanation, if a rational listener were to recognize that the speaker anticipated disbelief or disagreement, he[1] might even become suspicious of a claim that he otherwise would not have thought to question. As a result, reasons that *evidentially* count in favor of the conclusion can *pragmatically* count against it. I develop a model of the proposed "over-explanation" effect using the Rational Speech Act (RSA; Frank and Goodman, 2012) framework.

The experiment reported in the subsequent chapters investigates (1) whether listeners rationalize superfluous explanations by inferring that the speaker anticipated a disagreement, and (2) whether such an inference makes the explanation *less* convincing, not more. In a behavioral study, participants read a series of text-message conversations, in which a character justifies an action with zero or four reasons. In half the conversations, the justifications are provided out of the blue, and in the other half they are elicited discoursally to suppress the "over-explanation" inference. For each conversation, participants rated whether the speaker thought the listener would respond favorably or unfavorably, and whether the listener would actually respond favorably. The results of the study generally support the prediction of Queen Gertrude: participants indeed inferred that speakers had provided reasons out of the blue because they expected disagreement. Although the listeners' posterior beliefs were not credibly negatively impacted in either condition, the observed inference did significantly weaken the explanations' effectiveness. These findings contribute new data that can help clarify the relationship between trust and language processing.

[1]For clarity, the speaker is referred to as "she" and the listener as "he" going forward.

Chapter 2

Background

2.1 A rational theory of language processing

2.1.1 Rational speakers are efficient

"Information" in formal pragmatics is typically treated as a measure of logical strength. However, in certain computational and psycholinguistic traditions, it is a measure of *surprise*. On the latter accounts, any linguistic unit (word, phoneme, etc) is informative when it is surprising, and it is redundant when it is predictable (Levy and Jaeger, 2007). Following Shannon (1948), the information content ("surprisal") of a linguistic unit is quantified by its negative log-probability conditioned on the context:

$$I(u_i) = -\log P(u_i | u_0 \ldots u_{i-1})$$

This model describes the optimal rate of information ("bitrate") through a noisy channel with limited bandwidth. In order to maximize the amount of information that can be transmitted through the channel, a constant bitrate that approaches, but does not exceed, the channel's capacity must be maintained. Shannon proves that the most efficient coding scheme for a set of meanings M assigns $-\log P(m)$ bits to each $m \in M$. Less formally, longer signals are used for infrequently-occurring content, and shorter signals for frequently-occurring content, in order to maintain the optimal bitrate.

Natural languages are not optimal coding schemes, of course. However, it has long been of interest to psycholinguists whether speakers make "rational" usage of limited cognitive resources by constructing utterances efficiently (going back at least to Zipf,

1935). Over the past 20 years, numerous instantiations of the noisy-channel model have successfully captured patterns of signal reduction in low-surprisal contexts across various levels of linguistic structure. These include articulatory and durational phonetic reductions (Aylett and Turk, 2004; Bell et al, 2003; Pluymaekers et al, 2005), contractions (Piantadosi et al, 2011), lexical reductions (e.g. chimp/chimpanzee; Mahowald et al, 2013), complementizer omission (Jaeger, 2010), fragments (Bergen and Goodman, 2015; Lemke et al, 2021), and omission of discourse connectives (Asr and Demberg, 2020). All of these phenomena generally occur when the reduced or omitted content is contextually predictable.

"Rational" theories of language use– including Grice's (1975) maxim of Quantity– have previously come under fire because of their apparent inability to account for the logical redundancies that abound in naturalistic communication (Engelhardt et al, 2006; Heller et al, 2012). Studies show that speakers are willing to provide logically stronger (and seemingly less efficient) references than strictly required to identify a target:

(1) *Context: The display contains just one square.*
"Look at the **blue square!**"

However, recent work suggests that even logically redundant utterances conform to efficiency pressures, when "informativity" is construed probabilistically (Degen et al, 2020). Speakers are not "overinformative" at random. A longer utterance is worth its production cost if it helps reduce the listener's surprisal towards unpredictable material. For example, extra modification tends to occur in high-entropy contexts (e.g. references in complex visual displays; Rubio-Fernandez, 2016), and in contexts where misunderstanding is more frequent (e.g. learner-directed speech; Bergey et al, 2020; Tal et al, 2023). It also tends to occur with unpredictable attributes ("yellow strawberry" rather than "yellow banana"), and with uniquely identifying or less ambiguous attributes (e.g. color rather than size) that make the identity of the upcoming referent less surprising (Degen et al, 2020).

2.1.2 Rational listeners expect informativity.

If speakers usually expend more linguistic effort on less predictable content, we would expect listeners to interpret utterances with this production preference in mind. In fact, probabilistic pragmatic theories (e.g. Frank and Goodman, 2012) cite this as Grice's

core insight: comprehenders infer the state of the world based on the content of an utterance and the context in which it would be rational for the speaker to utter it.

Although many studies have shown that listeners tend to predict plausible content from the speaker (Bicknell et al, 2010), numerous others have shown a comprehension preference for utterances that are informative or newsworthy (in keeping with the previously described production preference for high-surprisal content; Hao et al, 2025; Rohde et al, 2021; Reksnes et al, 2024). For example, upon hearing the word "yellow", comprehenders expect the continuation to be something that can be yellow (a shirt), but isn't predictably yellow (a banana), unless in the presence of an atypical contrastive object (a green banana), as evidenced by both sentence completion and eye-tracking studies (Rohde and Rubio-Fernandez, 2022; Sedivy, 2003).

The observed expectation for informativity raises the question of whether listeners simply tolerate redundancies, or else derive pragmatic inferences to rationalize them instead. Although some experiments show that redundant references can be tolerated, more expensive propositional inefficiencies are increasingly shown to generate inferences (Kravtchenko and Demberg, 2022a; 2022b; Rees and Rohde, 2023; Ryzhova et al, 2023). In particular, assertion of seemingly predictable content causes listeners to revise their background beliefs, adjusting their model of the context to make the utterance situationally informative. For example, Kravtchenko and Demberg (2022a) demonstrate that script continuations, which are typically default inferences, can give rise to a "habituality" rationalization:

(2) Alice went to the store, and **she paid the cashier!**

Since "paying the cashier" would usually be trivially inferrable from world knowledge, listeners rationalize this type of utterance by assuming that Alice in particular doesn't typically pay.

2.1.3 Kravtchenko and Demberg's model

Kravtchenko and Demberg (2022b) model this inference using the RSA framework (Frank and Goodman, 2012; Goodman and Frank, 2016), which is frequently used to formalize recursive Gricean reasoning. The model developed in chapter 3 is adapted from this one, so in the interest of conciseness, further formal detail is reserved until then.

RSA models are grounded in a "literal listener" L_0, who predicts the true state of the world s, based on his prior beliefs and the literal semantics of the speaker's utterance u (and no additional pragmatic reasoning). K&D introduce to the basic formula a new parameter h, representing the habituality of the action (i.e. paying the cashier). The interpretation function $[\![u]\!](s)$ returns 1 if u is truth-conditionally consistent with world-state s, and 0 otherwise. The prior $P(s|h)$ represents the probability of the action taking place on this occasion, given standard world knowledge about its habituality.

$$P_{L_0}(s|u,h) \propto [\![u]\!](s) \cdot P(s|h)$$

Intuitively, for highly habitual actions (like script continuations), a literal listener will probably predict the correct world-state even if the activity is not explicitly mentioned, due to the high prior on h.

A pragmatic speaker S_1 selects her utterance (mentioning the action, or null) to minimize L_0's surprisal at s, while also minimizing the production cost C of the utterance [1]:

$$P_{S_1}(u|s,h,\alpha,\lambda,C) \propto P(u;\lambda,C) \cdot \exp(\alpha \log P_{L_0}(s|u,h))$$

Since L_0 is likely to (correctly) assume that a habitual action took place whether or not the speaker mentions it, S_1 can safely select the cheaper null utterance, without much risk of unclarity.

Finally the pragmatic listener L_1 jointly infers the world-state and habituality, based on prior world knowledge and expectations for S_1's behavior:

$$P_{L_1}(s,h|u) \propto P_{S_1}(u|s,h;\alpha,\lambda,C) \cdot P(s|h) \cdot P(h)$$

Since S_1 would prefer not to mention a habitual action when L_0 would assume it nonetheless, if S_1 does mention the action, L_1 is more likely to infer that, contra his prior world knowledge, this action is actually *not* habitual for this particular character. As such, L_1 predicts lower values of h when the action is mentioned, compared to the null utterance.

[1] The hyperparameters α and λ represent the respective weights of the informativity and economy/cost pressures

Crucially, though, the reduction of the habituality prior h induced by over-informativity does not affect the posterior probability of s. If the speaker says "she paid the cashier", the listener never believes otherwise, despite his inference that this occurrence is unusual. This is because L_0's interpretation function eliminates from consideration all possible worlds that contradict u, which is the standard dynamic effect attributed to assertion (Stalnaker, 1978).

The remainder of this paper proposes that information acquired by inference from reasons, just like information acquired by direct assertion, can also result in a reduction of the prior when offered redundantly. Furthermore, when the listener's degree of belief in s is determined by the probability of s in light of the new evidence and his background beliefs, and not just by recognition of the speaker's intention to convey s, the reduction of the prior *can* dampen the listener's posterior. This effect is the so-called "protest too much" inference.

2.2 From information to argumentation

2.2.1 Epistemic vigilance

When a speaker's communicative goal is to coordinate with the listener on some belief p, the most obvious strategy available to her is to make an assertion that uniquely specifies p among alternatives. Taking K&D's example, $u =$ "she paid the cashier" uniquely specifies paying the cashier; $u =$ NULL allows for the possibility of both paying and not paying. If the listener accepts the assertion of p, all $\neg p$ worlds are struck from the context set, so in probabilistic/ RSA terms, his surprisal at p is reduced to 0 (Stalnaker, 1978; see also Lassiter and Goodman, 2017).

Although this model is reasonable within the broader landscape of cooperative pragmatics, it is not obvious that actual listeners always, or even usually, assume their interlocutors are reliable, or that speakers expect listeners to make such an assumption. Sperber et al (2010) instead propose that language users are *epistemically vigilant*: we update our beliefs in response to testimony through some combination of trust in our interlocutors and the plausibility of the asserted content in the context of our other beliefs. Mercier (2020) produces extensive empirical evidence demonstrating that we are indeed far more discerning of the testimony we accept than previous work in psychology would suggest (e.g. the famous Gilbert et al, 1990).

Sperber et al's proposal departs from Grice on an essential point regarding informativity: when a rational speaker asserts p, she expects it to pass muster with a vigilant listener, not a fully trusting listener. Informing a listener that p is not as simple as specifying her own belief that p and expecting him to update his beliefs without further ado; p must also be sufficiently plausible within the listener's epistemic state. Speakers' preference to assert "believable" content is borne out empirically in studies of deception (Oey et al, 2023).

However, asserting p is not the only way speakers can inform listeners that p. While successful assertions conventionally reduce the surprisal of their own content to 0, they can affect the surprisal of numerous other contextually relevant propositions too: in particular, those that are inferentially/evidentially related to the asserted content (Winterstein, 2012; 2018). For example, successful assertion of "It's snowing" will maximize the listener's credence in (3), but presumably will also positively– though perhaps nonmaximally– affect his credence in (4).

(3) It's snowing.

(4) We should cancel school tomorrow.

There are, of course, many cases where a speaker would not assume a simple assertion of p would be admitted to the common ground unchallenged. For instance, p might be subjective (i.e. with judge-dependent truth conditions e.g. Stephenson, 2007; this is the focus of the experiment in chapter 4). It may otherwise be contextually implausible, or else it might contradict the listener's other public discourse commitments (Beltrama, 2018). In these cases, one strategy available to the speaker to raise the listener's credence in p is to produce a less challenging proposition q, which evidentially "counts in favor" of p:

(5) We should cancel school tomorrow, because it is snowing.

This strategy, of course, forms the basis of argumentation: the listener's surprisal at p is reduced due to the evidential support of q, and not due simple trust in the speaker's intention to convey p (as discussed by Sperber et al, this point is briefly conceded by Grice himself).

2.2.2 Previous approaches to argument strength

Although argumentation is a somewhat marginal case in the study of (cooperative) pragmatics, it is a central concern for philosophers and psychologists. The "counts in favor" relation has traditionally been treated categorically in the philosophical study of syllogistic argumentation: an argument is sound if the conclusion is a logical consequence of the reasons, and arguments that fail to meet this standard are considered fallacious even if they are psychologically compelling (see Mercier and Sperber, 2017 for an overview).

Natural language argumentation is hardly ever logically sound, yet it is intuitively the case that some informal arguments are more valid than others, even among instances of the same category of fallacy. For example, (6) and (7) are both instances of argumentation from ignorance, but (6) seems much more plausible (Oaksford and Hahn, 2004).

(6) The drug must be safe, because none of the clinical trials have failed.

(7) Ghosts must be real, because we have no evidence that they aren't.

To account for this variability, psychologists of reasoning have instead quantified an argument's strength in terms of its effectiveness in bringing about a desired belief. The latter approach lends itself to probabilistic, and specifically Bayesian models. A set of reasons argues in favor of the conclusion if the posterior probability of the conclusion conditioned on the reasons $P(C|R)$ is greater than the pre-utterance prior probability of the conclusion $P(C)$. The strength of a reason is quantified by the extent to which it raises the posterior probability, as computed through application of Bayes' Theorem (Hahn and Oaksford, 2007; Oaksford and Hahn, 2004).

$$\begin{aligned} P(C|R) &= \frac{P(R|C)P(C)}{P(R|C)P(C)P(R|\neg C)P(\neg C)} \\ &\propto P(R|C)P(C) \end{aligned}$$

An argument is *sufficient* if the posterior probability of the claim conditioned on the reasons exceeds some contextually determined threshold for acceptance (Godden and Zenker, 2018):

$$P(C|R) \geq \theta > P(C)$$

The sufficiency criterion allows the listener to accept a "good enough" naturalistic argument, even if the conclusion is not a fully logical consequence of the reasons.

This model has received modest attention from linguists studying argumentation. Within the framework of Argumentation within Language (*AwL*, e.g. Anscombre and Ducrot, 1983), a Bayesian semantics has been provided for a number of discourse connectives and particles which are proposed to operate on argumentative rather than truth-conditional meaning (Merin, 1999; Winterstein, 2012; 2018). Similarly, Cummins and Franke (2021) use the Bayesian model to quantify the effects of imprecise numeral interpretation on argumentative framing.

A tacit (and sometimes explicit, e.g. Hahn and Oaksford, 2007) assumption of a purely Bayesian argumentation theory is that, to a fully rational agent, the most convincing argument is the one that provides the strongest evidence from the reasons to the conclusion. Each additional positive reason can only monotonically increase the listener's credence by Bayesian conditionalization, so divergent behavior is attributed to failure of rationality. So, while this model does predict "diminishing returns" on additional reasons when the prior is already high, it does not predict that redundant reasons might actually count against the conclusion.

It is quite unclear, however, that listeners are always better off epistemically when more conclusive evidence for a claim is provided. Of course, the typical number of reasons provided to support any given conclusion is zero: most assertions are admitted on the basis of conversational norms (i.e. Grice, 1989). While there is considerable debate among epistemologists as to whether this tendency is justified or simply gullible (see Levy, 2021 for a recent contribution), it cannot be denied that language would be a highly *inefficient* means of acquiring knowledge if every proposition learned testimonially had to be fully, or even mostly, deducible from other known propositions.

2.2.3 The over-explanation inference

An alternate, information-theoretic analysis should (hopefully) present itself here. A rational speaker whose goal is to coordinate with the listener on p should select the most efficient utterance which sufficiently reduces his surprisal. In most cases, this is the bare assertion. But when common ground admissibility is questionable, she might

select a more expensive utterance that raises the epistemic or evidential standard of p. This point has been raised on numerous occasions in the formal literature, as a whole host of items marking truth (Romero and Han, 2004), certainty (Grice, 1969; von Fintel and Gillies, 2010), sincerity (Krifka, 2020), quality of evidence (Lai, 2012) etc. become available in contexts where the common ground admissibility of the prejacent is uncertain, and seem to run afoul of the Quantity maxim otherwise.

As such, a rational listener who encounters a detailed explanation for a given belief, assuming that the speaker is also behaving rationally, should infer that the speaker expected him to find that belief quite difficult to accept. As we will see shortly, this inference can be formalized as a reduction in the speaker's model of the listener's prior (similar to the reduction in K&D's habituality parameter). And since this is a case of argumentation, the listener's posterior belief is determined by conditionalization on the provided evidence, rather than simple acceptance of an assertion. Any adjustment of his background beliefs (i.e. to his own prior) *would* impact his posterior belief, unlike K&D's model of a strictly cooperative assertion.

Chapter 3

The Present Study

The objective of the present study is to determine whether pragmatic reasoning can affect the argument strength of an explanation. This chapter presents two models: one in which the listener updates his belief in the conclusion solely by conditionalizing on the reasons, and one in which the he evaluates evidence in light of the speaker's choice to provide it. The first model predicts that more evidence is always more persuasive. The second predicts that *sufficient* evidence is most persuasive, and that superfluous evidence can negatively impact the listener's posterior belief.

The space of explanations is constrained here to first-person justifications for actions, in the form "I did [action] because [reasons]". Justifications for actions argue principally in favor of the actions' utility or normative status (Kearns and Star, 2008), so argument strength is measured as the extent to which the addressee's judgment of the action is positively influenced by the reasons.

3.1 Notation

Following Achimova et al (2025), an agent's opinion state is modeled as a beta distribution defined on $O \in [0;1]$, a continuous variable representing a range of normative judgments from "strongly unfavorable" (0) to "strongly favorable (1). The mean/sample size beta parameterization is used, where the mean μ corresponds to the agent's absolute position, and the sample size ν to their "opinionatedness". For simplicity, ν is held constant at 3 here, representing fairly confident opinions, but see section 3.5 for further discussion. The notation $P(O;\mu)$ will be used to represent a beta distribution with mean

μ.

The listener's prior opinion state is modeled as some $P(O;\mu)$, where lower values of μ correspond to unfavorable priors, and higher values of μ correspond to favorable priors. If the listener doesn't reason pragmatically about justification, he will assume that the speaker's model of his prior is identical to his actual prior (no inference on μ). If he does reason pragmatically, he will have to infer which prior (hence, which μ) the speaker *expected* him to have. Some possible priors are shown in Figure 3.1

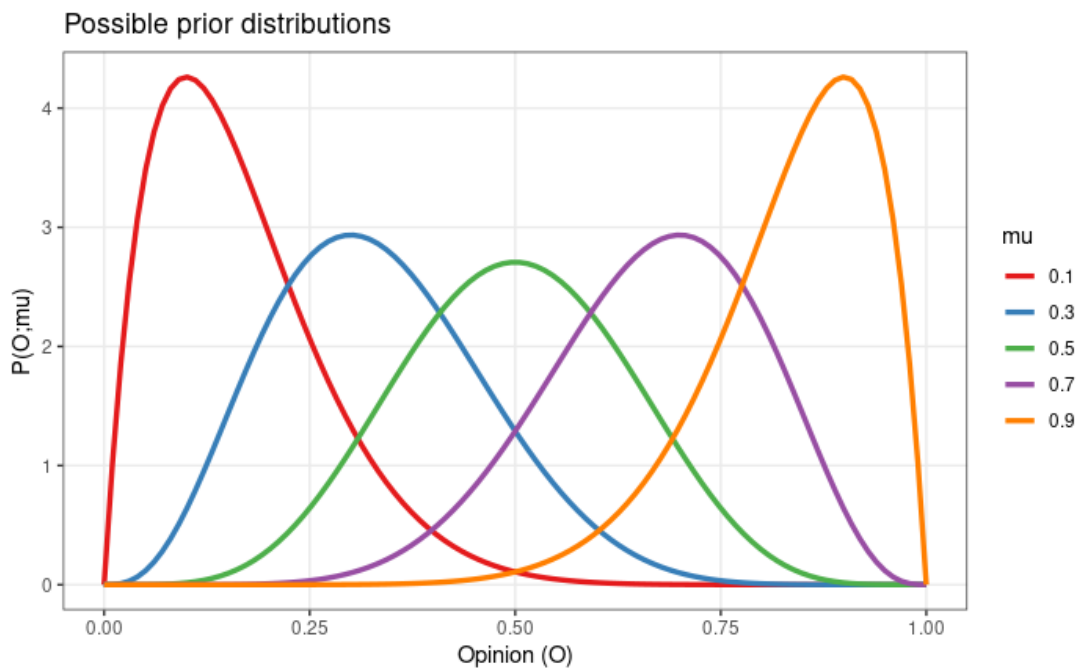


Figure 3.1: For lower values of the mean μ, the prior is skewed towards unfavorable values of O. For higher values of μ, the prior is skewed towards favorable values of O. Adapted from Achimova et al.

3.2 Research Questions

RQ1 (Pragmatic prior?): *Will the listener simply tolerate superfluous explanations? Or will he rationalize such utterances by inferring that the speaker expected him to have an unfavorable prior $P(O;\mu)$?*

RQ2 (Pragmatic posterior?): *Assume that the listener does indeed infer from over-explanation that the speaker expected an unfavorable prior (as per RQ1). Does that unfavorable prior negatively impact his own posterior belief $P(O|R)$?*

3.3 Hypothesis 1: More is more.

The first model predicts that listeners do not reason pragmatically about the amount of explanation a speaker chooses to provide. Regarding RQ1, as already stated, whether or not the speaker gives a long explanation, he will assume that her model of his prior is simply his actual prior. As such, the amount of explanation he receives should not affect his estimation of the speaker's model of his prior.

Following the simple Bayesian account of argument strength, the listener's posterior belief can be computed with a straightforward application of Bayes' Theorem:

$$P_L(O|R;\mu) \propto P(R|O) \cdot P(O;\mu) \tag{3.3.1}$$

The listener assigns some probability to each possible value of O according to his own prior $P(O;\mu)$ and (if reasons are given) the likelihood $P(R|O)$. For ease of modeling, a linear likelihood function of $P(R|O) = O$ is used, since positive reasons are more likely to be true when O is more favorable.

As for RQ2, if no reasons are given, the posterior reproduces the prior. If many reasons are given, the posterior always shifts the probability mass to higher values of O in comparison to the prior, indicating more positive opinions (although for high priors, the increase is less significant, see Figure 3.2).

3.4 Hypothesis 2: Less is more.

The second hypothesis predicts that listeners do reason pragmatically about the amount of explanation a speaker provides. The inference is modeled here using the RSA framework.

3.4.1 Literal listener

The pragmatic model is adapted from Kravtchenko and Demberg's (2022b) RSA setup. As is standard for RSA models, the recursive social reasoning is grounded in a "literal listener", who determines the state of the world based solely on the content of the utterance and prior beliefs about the world; this hypothetical listener makes no pragmatic inferences.

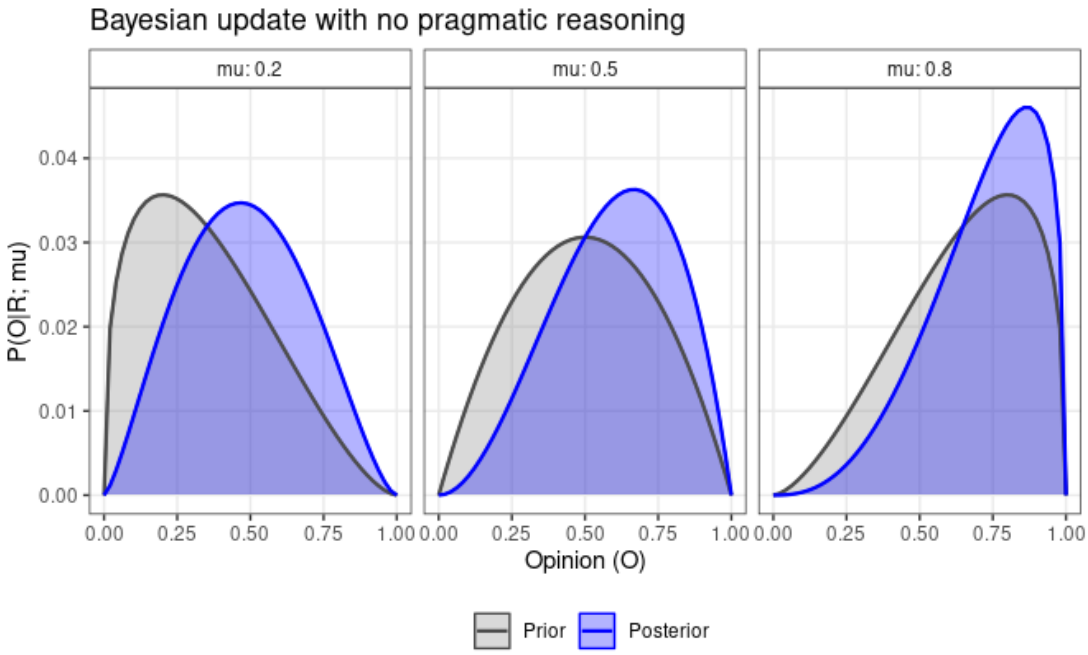


Figure 3.2: For all priors, an explanation increases the probability of the listener arriving at a more favorable opinion. However, the this effect is stronger for low priors, and weaker for high priors.

$$P_{L_0}(O|R;\mu) \propto P(R|O) \cdot P(O;\mu) \tag{3.4.1}$$

However, this model departs from K&D's (as well as any standard RSA implementation) since L_0's usual interpretation function $[\![u]\!](s)$ is replaced with a basic Bayesian likelihood. The purpose of the interpretation function is to model the effects of a direct assertion on an information state: that is, eliminating all worlds that are incompatible with the asserted content. This model, however, is intended to describe the informational effects of an inference from evidence: that is, how the listener's credence in O is raised by learning that R, as standardly modeled by Bayes' Theorem.

For example, in K&D's setup, assertion of "My aunt is sick" raises the listener's credence in (1), the literal content of the assertion itself, to 1. This setup is intended to show how (1) evidentially affects his belief in (2).

(1) My aunt is sick.

(2) It's good/favorable for me to visit her this afternoon.

As with K&D's habituality parameter h, the literal listener here does not infer the prior mean μ; this value represents what he is expected to already believe, separately from the utterance. When the literal listener's prior opinion of the action is unfavorable (lower μ), providing more reasons will meaningfully improve it (more probability mass on higher values of O, compared to when R is null). On the other hand, for an action which already has a favorable prior, the explanation wouldn't significantly improve the literal listener's opinion. However, at least some improvement is always observed. As such, the literal listener's behavior is identical to the predictions of Hypothesis 1.

3.4.2 Pragmatic speaker

The pragmatic speaker's goal is to ensure that the literal listener arrives at a favorable opinion O, while minimizing the production cost of her utterance. Here, a "favorable opinion" is taken to be when the literal listener's opinion exceeds some threshold value (e.g. Godden and Zenker, 2018). The probability mass of P_{L_0} in excess of a threshold θ is computed as follows:

$$P_{L_0}(O > \theta | R; \mu) = \int_{\theta}^{1} P_{L_0}(O|R;\mu) dO \tag{3.4.2}$$

The utility of an explanation for the speaker, then, is its inferential informativity minus its cost. As with standard RSA, informativity is quantified as the literal listener's negative post-utterance surprisal towards the target belief. The most informative utterance here is the one which makes the literal listener most confident that the speaker's action was a good one:

$$\mathbb{U}_{S_1}(R;\theta,\mu) = \underbrace{\log P_{L_0}(O > \theta|R;\mu)}_{\textbf{-surprisal}} - \text{cost(R)} \tag{3.4.3}$$

The pragmatic speaker selects an utterance (reasons or no reasons) by sampling from the softmax distribution over her utility. The parameter α is the softmax temperature which represents the extent to which the speaker behaves "rationally" (with higher α, the speaker maximizes her utility, with lower α, she selects randomly. Here $\alpha = 5$).

$$P_{S_1}(R|\theta,\mu) = \text{softmax}(\alpha \cdot \mathbb{U}_{S_1}(R;\theta,\mu)) \tag{3.4.4}$$

Since S_1 trades off utterance cost with her communicative goal to induce a favorable opinion, we would expect her to choose the costly R only when it would have a meaningful impact on the listener's expected judgment. When L_0's expected prior is low, the explanation is necessary to obtain the threshold, but when his prior is high, the simple assertion is nearly as effective and far more economical. So the speaker's utterance probability for the explanation is higher for low values of μ, and lower for high values of μ (Figure 3.3).

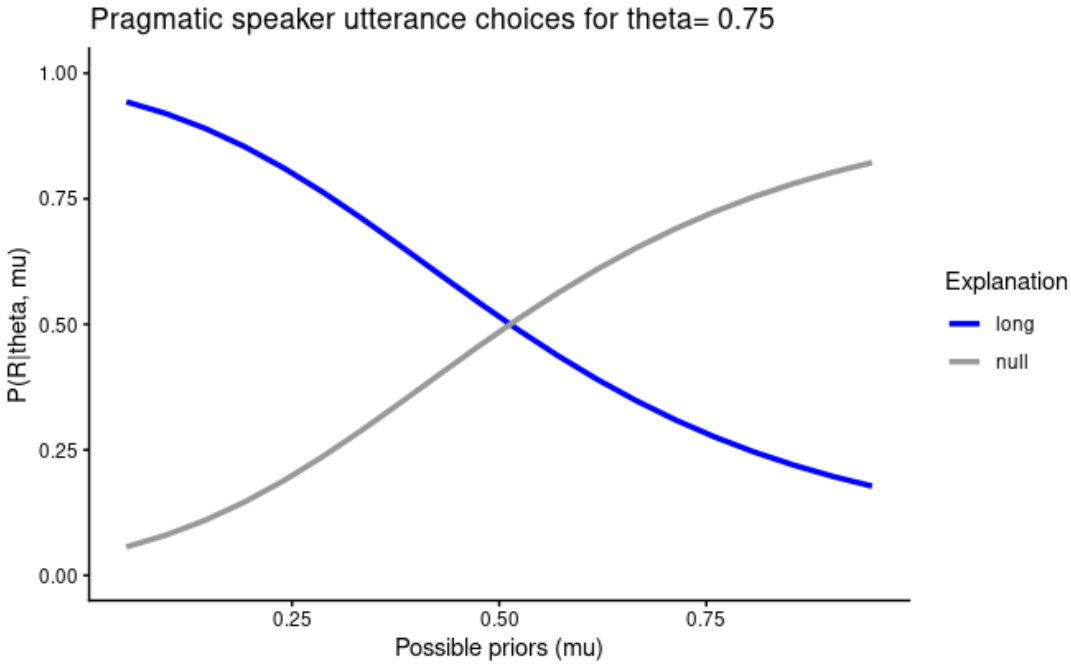


Figure 3.3: Speaker production probabilities for a threshold of 0.75. With low expected priors, more explanations are predicted. With high priors, fewer explanations are predicted.

3.4.3 Pragmatic listener

As with K&D's model, the pragmatic listener L_1 first infers which world state the speaker believes to be true, under the assumption that she has selected her utterance rationally (maximizing her utility). He performs a joint inference on two variables: the speaker's desired threshold θ, and the prior opinion that she expected L_0 to have, μ, in light of his priors over θ and μ and the likelihood of the speaker producing the explanation R given θ, μ.

$$P_{L_1}(\theta,\mu|R) \propto P_{S_1}(R|\theta,\mu) \cdot P(\theta) \cdot P(\mu) \tag{3.4.5}$$

The probability of each value of the speaker's expected prior μ can be calculated by marginalizing over θ, see also Figure 3.4.

$$P_{L_1}(\mu|R) = \int_0^1 P_{L_1}(\theta,\mu|R)d\theta \tag{3.4.6}$$

Since S_1 tends to select long explanations for unfavorable priors, and null explanations for favorable priors, L_1 tends to predict lower values of μ after hearing a long explanation, and higher values of μ with no explanation. With respect to RQ1, this model predicts that listeners do rationalize over-explanation by inferring that the speaker predicted an unfavorable prior.

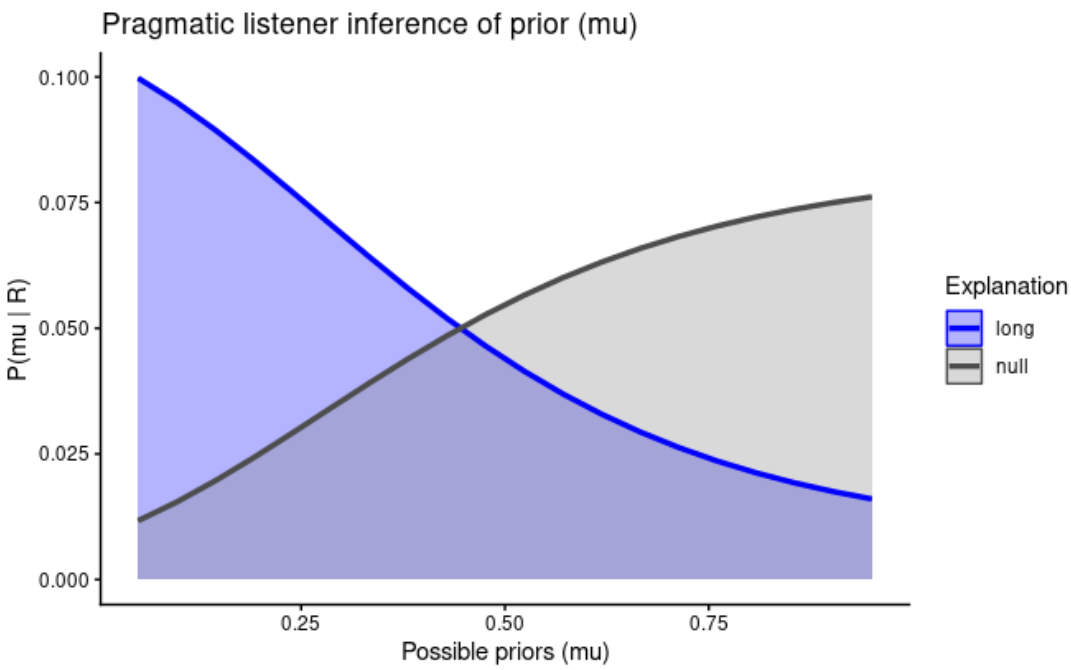


Figure 3.4: L1 infers S1's model of L0's prior based on her utterance. He tends to predict lower values of the prior when he hears a long explanation. Uniform threshold prior $P(\theta)$ and hyperprior $P(\mu)$ are used here.

Just like L_0, L_1 also estimates the value of O by conditioning the prior $P(O;\mu)$ on the reasons R. But since L_1 reasons pragmatically, he infers the prior based on how S_1 is likely to behave, as well as his own world knowledge. He arrives at his overall posterior belief by taking a weighted average of the posteriors obtained by conditionalization from each possible prior $P(O;\mu)$:

$$P_{L_1}(O|R) \propto \int_0^1 \underbrace{P(R|O) \cdot P(O;\mu)}_{\textbf{Conditionalization from prior}} \cdot \underbrace{P_{L_1}(\mu|R)}_{\textbf{Probability of that prior}} \, d\mu \tag{3.4.7}$$

The pragmatic model also predicts a positive answer to RQ2, whether over-explanation can backfire and produce a lower posterior belief in the reasons case than in the no-reasons case. If L_1 originally had an unfavorable opinion (i.e. an unfavorable hyperprior $P(\mu)$), then his posterior belief is likely to still somewhat increase with the long explanation. Intuitively, he would *expect* the speaker to justify her action if own opinion is unfavorable. However, if he originally held a favorable opinion (favorable hyperprior $P(\mu)$), his posterior belief can decrease with the long explanation due to his inference that the speaker expected a worse prior than he himself expected (Figure 3.5).

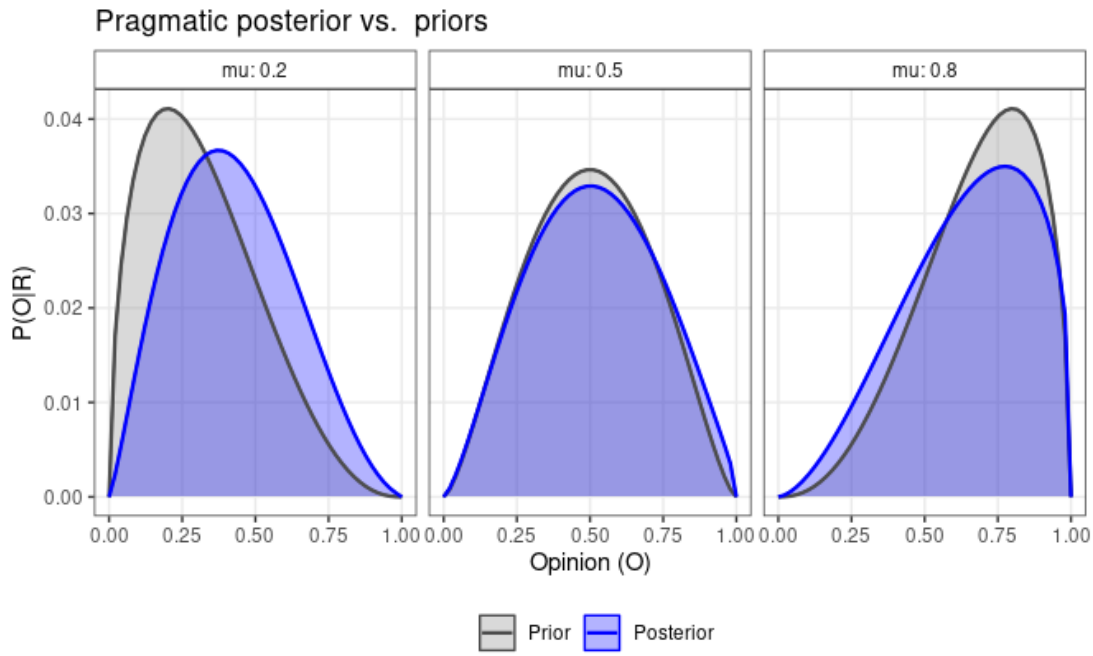


Figure 3.5: For a low hyperprior, providing an explanation increases the posterior. For a high hyperprior, the explanation instead decreases the posterior.

3.5 Note on sample size

As previously mentioned, it makes sense that a listener would expect to hear some justification if his own pre-utterance belief is unfavorable. If his actual prior is low, it would not be surprising to him that the listener also expected an unfavorable prior. This

is why the backfire effect is not enough to result in a net decrease of his posterior for unfavorable hyperpriors.

It also seems logical that the listener would expect to hear justification if it's not obvious whether the action is *usually* good or bad– that is, when his prior is underinformed, rather than unfavorable. This corresponds to the previously mentioned sample-size parameter ν. Indeed, repeating the same inference but with reduced $\nu = 2$, we can see that the backfire effect is much weaker even for high values of μ.

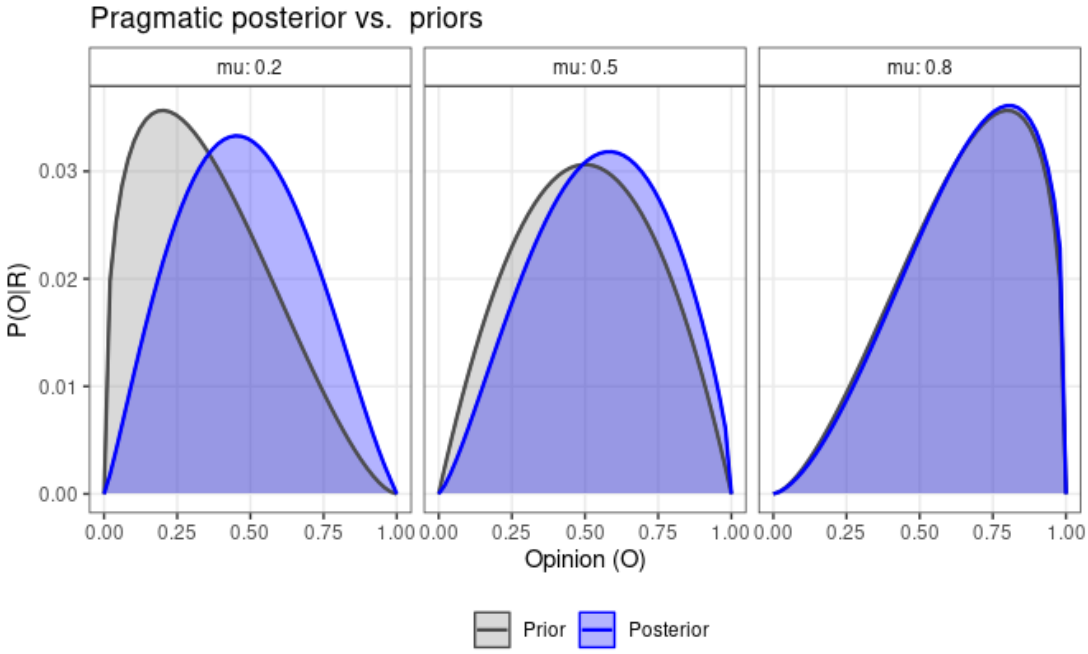


Figure 3.6: The over-explanation backfire is suppressed for underinformed priors.

Chapter 4

Methods

4.1 Participants

60 participants were recruited through Prolific. All participants self-identified as native speakers of English, and were residents of the United Kingdom or United States. An additional 6 participants were recruited for a shorter pilot study (with the free-text question omitted); their data is not included in the subsequent analysis. The survey took a median time of 17 minutes, 52 seconds to complete, and participants were compensated £5.00. All participants in the main study passed the attention checks, so they were all included in the analysis.

4.2 Study Design

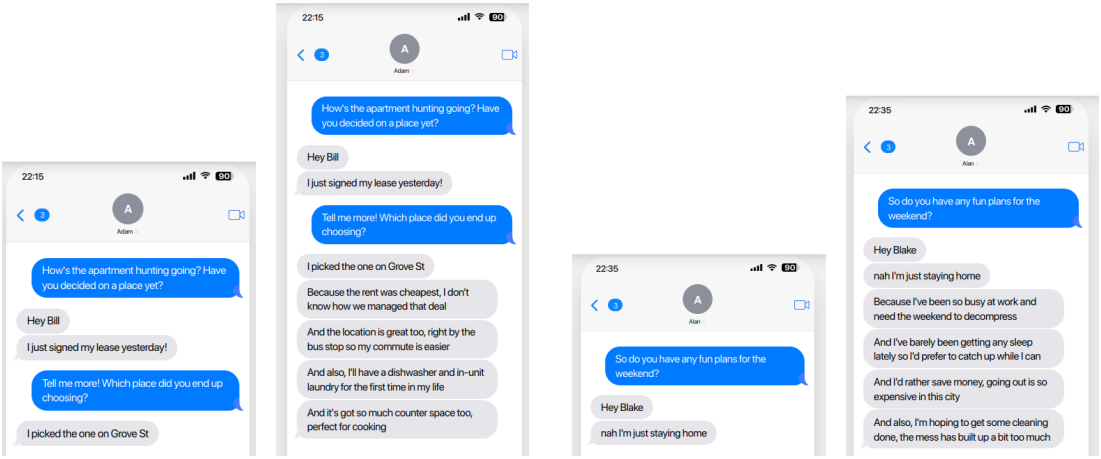
The experiment was constructed using a 2x2 Latin square design, crossing expectation (**reasonsUnexpected** vs **reasonsExpected**) with reasons (**noReasonsGiven** vs **reasons-Given**. Items were sorted into two lists, with 4 trials in each of the 4 conditions. Each item appeared in only one condition per list. Participants also saw 4 attention checks, for a total of 20 trials.

4.3 Materials

Prior work has shown that comprehenders have a greater expectation for informativity when the utterance is produced by a salient speaker (for example, a named character), so redundancies are more easily perceived as deliberate (Reksnes et al, 2024). As such, the

stimuli for this study were presented as text message conversations between two named characters, adapting the formatting of Wallbridge et al (2021). To avoid confusion, the explainer's name always began with A, and the addressee's name always began with B.

Since the research questions target comprehenders' inferences about speakers' production choices (rather than speakers' production choices themselves), participants were made to assume the role of the addressee rather than the explainer. Therefore, the explanation always came from the "incoming"/grey bubble character, and both rating scale questions concerned the opinion of the "outgoing"/blue bubble character.



Sample **reasonsExpected** stimuli

Sample **reasonsUnexpected** stimuli

Figure 4.1: Sample stimuli in **noReasonsGiven** and **reasonsGiven** conditions.

For the **reasonsUnexpected** class of items, character A describes an "ordinary" altruistic or innocuous behavior (e.g. cleaning the kitchen for a roommate or studying in the library), where the participants were expected to have favorable priors. The reasons (when provided) were offered unprompted.

For the **reasonsExpected** class of items, character A describes a behavior that typically involves a deliberative process where multiple reasons would be considered (e.g. moving to a new city or making an expensive purchase), in response to a "which one"/"tell me more"-type question from character B. This was meant to create a context where the participants might expect either underinformed or unfavorable priors, which are expected to suppress the inference.

In the **noReasonsGiven** condition, character A simply mentions the behavior.

In the **reasonsGiven** condition, character A provides four reasons. The content of the explanation was not held constant across item types, but was instead tailored to provide

strong/sufficient justification for each individual behavior (to avoid confounding with the "weak evidence" effect e.g. Barnett et al, 2022; as well as to maintain the naturalness of the dialogues). All explanations were approximately the same total length (although some variability was introduced, again to maintain naturalism).

Realism of the stimuli was prioritized so that the participants might have actual intuitions about why a justification would or would not be provided in a given circumstance, following prior work on pragmatic inferences generated by strategic language. For example, while a number of studies have claimed that participants nearly-uniformly do not make lie judgments on implicit meaning, Weigmann (2023) shows that such judgments are simply suppressed in odd situations where the characters' motives for deception are unintuitive.

In all target trials, the messages cut off before character B responds. An additional 4 attention checks were constructed, where character B does react to character A's message with explicit approval or disapproval (2 items for each), so the participant could simply report B's perspective without inference.

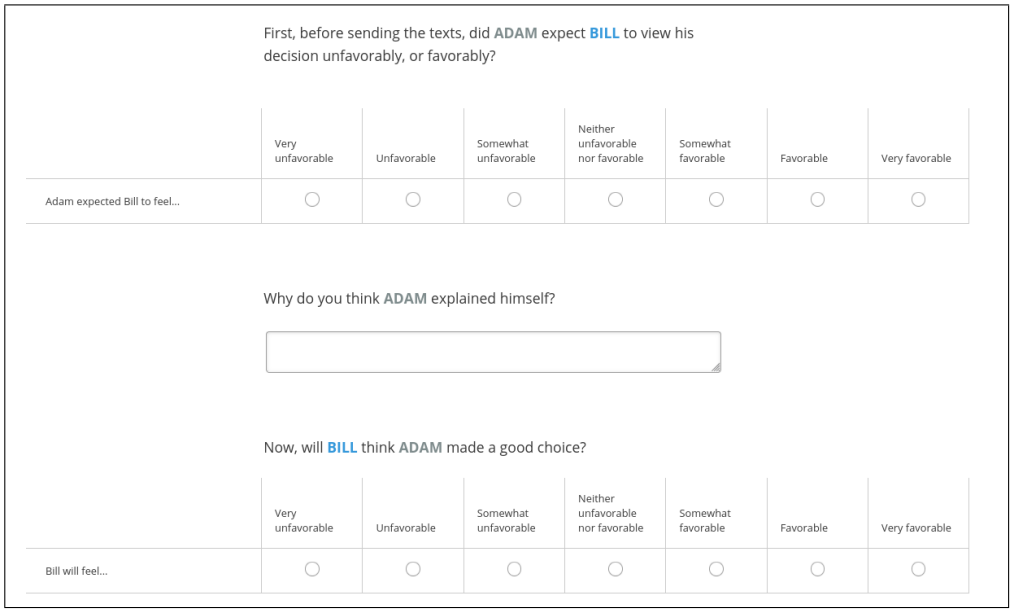


Figure 4.2: Sample questions in the **reasonsGiven** condition.

Each target trial contained two or three questions. First, participants were asked to rate the following on a 7-point Likert scale from "strongly unfavorable" to "strongly favorable":

- First, before sending the texts, did A expect B to view her decision unfavorably,

or favorably? A expected B to feel…

- Now, will B think A made a good choice? B will feel…

The first rating was intended to target RQ1: whether the listener would think the speaker had explained her decision because she expected him to disagree with it. The second rating was intended to target RQ2: whether the listener's recognition of that expectation could counteract the effectiveness of the reasons in influencing his beliefs. In other words, the first question identifies whether the "pragmatic prior" is lower (less favorable) in the **reasonsGiven** condition than in the **noReasonsGiven** condition, and whether that effect is moderated by expectedness of the explanation. The second question identifies whether the lower prior produced by the explanation would then yield a reduction in *post-utterance belief* (again, a reduction in favorability rating from the **noReasonsGiven** condition to the **reasonsGiven** condition– and again, whether that effect is moderated by expectedness of the explanation).

Participants were also asked to provide a free-text response to the question "Why do you think A explained herself?" for items in the **reasonsGiven** condition, to see if participants preferred an alternative rationalization for the over-informativity besides the hypothesized judgments measured by the Likert ratings. This additional question was included in the main experiment to provide more interpretable data for analysis, after conducting a preliminary pilot study which only included the Likert ratings.

4.4 Procedure

The experiment was hosted online on Qualtrics. Before proceeding to the survey, participants were asked to read an information sheet and consent to participation (approved by the University of Edinburgh LEL Research Ethics Board) and to confirm their Prolific ID and native-speaker status. The participants were then instructed to read each set of messages closely and reason about the characters' motives for sending them. Using the Qualtrics survey randomizer, each participant was assigned to one of the two stimulus lists, and saw the items in random order. Each trial was presented on a separate page, and participants were required to answer each of the questions before clicking to advance. At the end of the survey, participants completed a "debrief", where they were asked to provide feedback and guess what the experiment was about.

Chapter 5

Results

5.1 Model selection

Data analysis for the Likert rating questions was conducted using a Bayesian cumulative logit model. Although linear mixed-effects models are standard practice in psychology for 7-point Likert scales (assumption of equidistant category boundaries is more acceptable with high-granularity ordinal scales), the Bayesian cumulative model is still best practice for ordinal data since it makes no assumptions regarding the spacing of the categories (Bürkner and Vuorre, 2019). The model was implemented using the R (R Core Team, 2025) package `brms` (Bürkner, 2017) with default priors. The fixed effects included REASONS (**noReasonsGiven** vs **reasonsGiven**), EXPECTATION (**reasonsUnexpected** vs **reasonsExpected**), and their interaction. To account for participant and item-level variability, random effects for PARTICIPANT and ITEM were also included.

5.2 Do listeners rationalize over-explanation?

The first Likert rating question (*Before sending the texts, did A expect B to view her decision unfavorably, or favorably?*) measures the pragmatic prior. It was intended to target RQ1: whether listeners would attempt to rationalize a speaker's decision to (over)explain her actions, by inferring that she predicted a disagreement. Responses are plotted in Figure 5.1.

Treating **noReasonsGiven** and **reasonsUnexpected** as the baseline conditions, the

model identified a credible main effect of REASONS (β = -0.77, 95% CI -1.09, -0.43), indicating a reduction in favorability rating when reasons were given. This effect was moderated by a credible interaction of REASONS with EXPECTATION (β = 0.83, 95% CI 0.35, 1.30), which reversed the main effect entirely when the reasons were expected. In other words, participants rated the speaker's estimation of the listener's opinion credibly lower when reasons were *unexpectedly* provided. Participants did not rate the speaker's estimation of the listener's opinion credibly lower when the reasons were *expectedly* provided. There was also no credible main effect of EXPECTATION, indicating that **reasonsExpected** items were not consistently rated higher or lower in favorability than **reasonsUnexpected** items in the **noReasonsGiven** condition. Results of the models are summarized in Table 5.1.

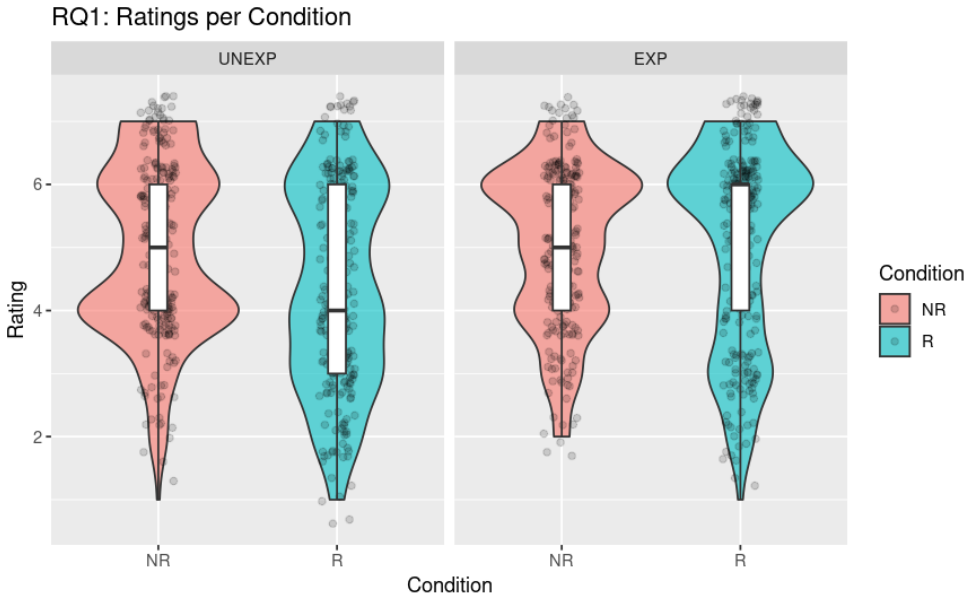


Figure 5.1: Responses to RQ1. Prior favorability ratings reduced when reasons were provided unexpectedly.

These results are in accordance with the predictions of the pragmatic hypothesis: when many reasons are unexpectedly provided, the listener rationalizes the speaker's decision to provide them. The listener infers that the speaker expected him to disapprove of her action– that he would need to be *convinced* to agree with her. The result is a reduced "pragmatic prior" favorability rating. However, in a context where many reasons are appropriate, the inference is suppressed. The literal hypothesis would instead predict no effect for either item type (expected or unexpected), since the listener does not reason about the speaker's choice to explain herself in any case.

Predictor	Estimate	Std. Error	95% CI Lower	95% CI Upper
conditionR	**-0.77**	**0.17**	**-1.09**	**-0.43**
typeEXP	0.19	0.66	-1.12	1.52
conditionR:typeEXP	**0.83**	**0.24**	**0.35**	**1.30**

Table 5.1: Log-odds coefficients from the Bayesian ordinal model of RQ1. A credible main effect of REASONS, moderated by a credible interaction with EXPECTATION, is observed.

5.3 Does over-explanation affect argument strength?

The second Likert scale question (*Now, will B think A made a good choice?*) measures the listener's posterior favorability rating. It was intended to target RQ2: whether higher-order inferences about *why the speaker felt an explanation was necessary* would mitigate that explanation's effectiveness in persuading the listener. Responses are plotted in Figure 5.2.

Once again treating **noReasonsGiven**, **reasonsUnexpected** as the baseline conditions, there was no credible main effect of REASONS for **reasonsUnexpected** items. However, the model identified a credible interaction of REASONS with EXPECTATION ($\beta = 0.75$, 95% CI 0.28, 1.22), meaning that the posterior favorability ratings improved when the reasons were *expected*. Although unexpected reasons had no effect on the listener's post-utterance favorability ratings, expected reasons significantly improved the listener's rating. There was still no credible main effect of EXPECTATION, indicating that **reasonsExpected** items were not consistently rated higher or lower than **reasonsUnexpected** items in the **noReasonsGiven** condition. Results of the models are summarized in Table 5.2.

Predictor	Estimate	Std. Error	95% CI Lower	95% CI Upper
conditionR	0.15	0.17	-0.20	0.50
typeEXP	0.25	0.76	-1.25	1.79
conditionR:typeEXP	**0.75**	**0.24**	**0.28**	**1.22**

Table 5.2: Log-odds coefficients from the Bayesian ordinal model of RQ2. A credible interaction of REASONS and EXPECTATION is observed.

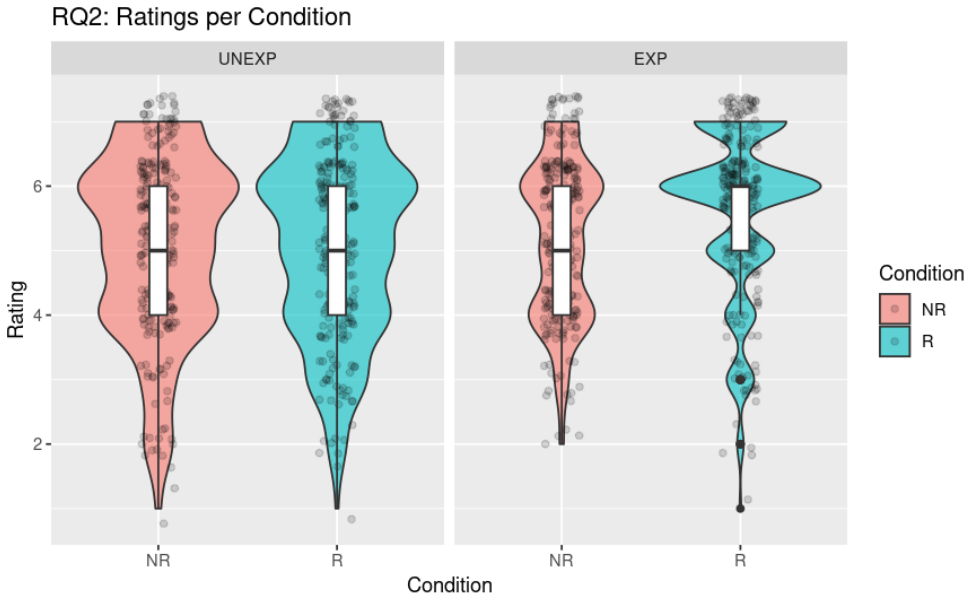


Figure 5.2: Responses to RQ2. Posterior favorability ratings increased with reasons only when the reasons were expected.

There was no credible *reduction* in listeners' posterior favorability rating in the **reasonsGiven** condition, suggesting that the higher-order inference about why the speaker wanted to explain herself was insufficient to fully counteract the argumentative strength of the reasons. However, these results still support a pragmatic analysis. It should be noted that there was no credible effect of EXPECTATION for items in the **noReasonsGiven** condition: that is, there was no significant difference between baseline approval ratings for the **reasonsExpected** and **reasonsUnexpected** items. However, the explanations improved post-utterance listener favorability ratings only for **reasonsExpected** items; they had no credible effect for **reasonsUnexpected** items. This suggests that, although there was similar "room for improvement" in both conditions (i.e. it was not the case that listeners *already* maximally approved of the **reasonsUnexpected** items), listeners were only persuaded in the **reasonsExpected** condition. If this is indeed the case, the argumentative strength of the reasons could have been dampened by the reduction of the pragmatic prior identified in Q1, as predicted by the pragmatic hypothesis.

To corroborate this analysis, a new model was constructed to examine the difference between participants' responses to the two questions (effectively Q2-Q1, but maintaining the ordinality of the data); that is, to measure the difference between the pragmatic prior (Q1 rating) and the listener's post-utterance belief (Q2 rating). The model included

Predictor	Estimate	Std. Error	95% CI Lower	95% CI Upper
condR	-0.77	0.17	-1.11	-0.44
Q2	0.19	0.17	-0.15	0.52
typeEXP	0.24	0.68	-1.10	1.60
condR:Q2	**0.95**	**0.24**	**0.48**	**1.44**
condR:typeEXP	0.86	0.24	0.40	1.32
Q2:typeEXP	-0.03	0.24	-0.51	0.44
condR:Q2:typeEXP	-0.17	0.34	-0.87	0.49

Table 5.3: Log-odds coefficients from the three-way model. A credible interaction of QUESTION and REASONS is observed.

an additional fixed effect of QUESTION number (and its interactions), with all other parameters maintained from the previous models. The new model found no credible main effect of QUESTION, meaning that the pragmatic prior (Q1 rating) and posterior (Q2 rating) were about the same when no reasons were provided. There was a credible interaction of QUESTION and REASONS: the posterior (Q2) was credibly higher than the pragmatic prior (Q1) when reasons were provided. Finally, there was no credible three-way interaction between QUESTION, REASONS, and EXPECTATION, meaning that the increase in favorability from the pragmatic prior to the listener's posterior belief was similar for both **reasonsUnexpected** and **reasonsExpected** items when reasons were provided.

Summing up, then: the Q1 model found that the that the pragmatic prior was reduced in the **reasonsGiven** condition for **reasonsUnexpected** items, but it was not reduced for the **reasonsExpected** items. The three-way model found that the *difference* between the pragmatic prior and the posterior favorability was equivalent for both **reasonsUnexpected** and **reasonsExpected** items. Therefore, items in the **reasonsExpected** set saw a net improvement from the **noReasonsGiven** condition to the **reasonsGiven** condition, whereas the items in the **reasonsUnexpected** set simply recovered from the reduced pragmatic prior and returned to "baseline", as per the findings of the Q2 model.

5.4 Notes on the free-text data

Although full quantitative analysis of the free-text responses is beyond the scope of this paper, some high-level analysis of the results that emerged are discussed in the

following chapter.

5.5 Notes on the debrief

At the end of the survey, participants were asked to report what they thought experiment was investigating. The vast majority of respondents did not mention explanation/justification at all. Many reported that they could not even guess what the experiment was about, and several others suggested that it might be testing how the text-message medium affects tone recognition. These results indicate that participants did not trivially identify the research questions and attempt to give "correct" responses, which provides validation for the experiment design.

Chapter 6

General Discussion

6.1 Summing up

The results of the behavioral study show that comprehenders do consider speakers' production preferences for communicative efficiency when evaluating explanations. They do not increase their belief in a proposition just in accordance with the weight of evidence provided in its favor. Instead, they identify when more reasons are provided than expected, and derive inferences about what prior beliefs the speaker predicted them to have held, in order to make an argument worthwhile. These pragmatic inferences seem to affect the strength of the argument: when more reasons are given than expected, the argument fails to influence the listener's posterior beliefs in the desired direction, resulting in a sense of "the lady doth protest too much." But when reasons are contextually appropriate, they reliably influence the interlocutor's beliefs in the desired direction.

6.2 Limitations and future work

6.2.1 Non-monotonicity

Although the present findings are in support of a pragmatic analysis of argument strength, there is no evidence for true non-monotonicity. While it seems that the reduced prior did, in fact, weaken the argument for the **reasonsUnexpected** items in the **reasonsGiven** condition in comparison to the **reasonsExpected** items, it was not enough to produce post-utterance favorability ratings lower than in the **noReasonsGiven**

condition. Impressionistic evaluation of the free-text data suggests some potential avenues for further work: some participants commented that it was difficult to determine whether the character made a good decision when no reasons at all were given, while others commented that many of the explanations seemed excessive or unnecessary (as designed). Non-monotonicity might be more easily identified, then, if a third condition with just one or two reasons was added to the experiment. If participants are unconvinced when no reasons are given, and suspicious when too many are given, perhaps the optimal number is somewhere between the two extremes. A reduction in argument strength might then be observed between the "few reasons" and "many reasons" conditions.

6.2.2 Discourse vs subject-matter manipulation

Another useful qualitative finding from the free-text responses was that the discourse manipulation seemed much more effective in dismissing the "defensive" interpretation for **reasonsExpected** items than the subject-matter manipulation. Many participants commented that the addressee seemed like they wanted to know more, although some did also comment on the action itself as well. Furthermore, as mentioned, there was no credible difference between the ratings in the **reasonsExpected** and **reasonsUnexpected** conditions when no reasons were provided. Taken together, these findings seem to suggest that a low sample size parameter ν (underinformed or "curious" prior) can suppress the inference even when the mean μ is not particularly unfavorable. A more precise RSA design, then, would involve a three-way inference on both beta parameters (μ, ν) and the threshold at the pragmatic listener step. In fact, it seems likely that the backfire effect might arise even for low μ if the listener is "set in his ways" (high ν). Future work might also necessitate a norming study to get empirical estimates for μ and ν hyperpriors.

6.2.3 Different types of normative judgment

This study was primarily concerned with the absolute quantity of reasons. However, the free-text results also suggest that the *type* of reason might have an effect on the inferences that arise. The scenarios used in this experiment involved justifications for actions, and the utility of an action might be influenced by a number of different judgments (desirability, permissibility, obligation, etc). All of these were collapsed into a single judgment of "favorability". The free-text responses suggest that participants

made more fine-grained judgments based on the content of the reasons: for example, participants identified that a certain character wanted to make a charitable donation (despite having no obligation to do so) based on the content of the reasons, which subsequently generated an impression of "virtue signaling". They identified that another character had to go to the library (despite not wanting to), which repeatedly generated the inference that he was avoiding spending time with the listener, by making the studying seem more urgent than it actually was. Further work might consider whether the type and typicality of the reasons provided can affect the extent to which over-explanation can backfire, and what specific inferences are made in response (in the same way that the type and typicality of modifiers affects the informativity inferences they generate; Degen et al, 2020; Sedivy, 2003).

6.3 Concluding Remarks

Ultimately, this study constitutes a preliminary effort to bridge the gap between cooperative pragmatics and argumentation, which is frequently envisioned as an adversarial mode of communication (Merin, 1999). If listeners in the real world are expected to reject implausible testimony, then it makes sense that rational speakers hoping to ensure belief coordination should construct their utterances to be credible, rather than assuming their assertions will be accepted by default. On the other hand, if listeners are generally inclined to accept testimony on the basis of conversational norms, then rational speakers hoping to maximize the rate of information exchange should avoid producing unnecessary justification. It is established by now that there must be some correlation between language processing and truth judgment (Sperber et al, 2010), yet the nature of this correspondence is still quite opaque. Future work on this topic, I hope, can help shed new light on the relationship between language and belief.

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